Technical Case Study

Position: Senior Data Quality Engineer

Objective

The purpose of this case study is to assess the candidate's practical ability to design and implement data quality controls within a modern enterprise data platform. The assessment focuses on data profiling, DQ rule design, automated controls, reconciliation, exception handling, monitoring, root-cause analysis, and implementation across Medallion Architecture.

Scenario

The organization is implementing an enterprise data platform using a Medallion Architecture (Bronze → Silver → Gold). Student enrollment data is received from multiple higher-education institutions and loaded into the platform.

Student: Student_ID, Emirates_ID, Full_Name, Date_of_Birth, Gender, Nationality_Code, Institution_Code, Created_Date, Updated_Date

Enrollment: Enrollment_ID, Student_ID, Institution_Code, Program_Code, Academic_Year, Enrollment_Status, Enrollment_Date, Graduation_Date

Institution: Institution_Code, Institution_Name, Institution_Status

Program: Program_Code, Institution_Code, Program_Name, Program_Status

The following data-quality issues have been observed:

- Missing Student IDs
- Duplicate student records
- Invalid institution or program codes
- Enrollment records referencing students that do not exist
- Invalid or inconsistent dates
- Different representations of the same reference values
- Differences between source and target record counts
- Late-arriving records

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Candidate Tasks

1. Data Profiling

Explain how you would profile the datasets before defining the DQ rules. Identify the main checks, statistics, and metrics you would capture.

2. Data Quality Rules

Define at least 8–10 DQ rules covering different dimensions such as completeness, validity, uniqueness, consistency, referential integrity, and timeliness. For each rule, specify the DQ dimension, target layer, severity, and expected action.

3. Medallion Architecture Implementation

Explain which DQ controls you would implement in Bronze, Silver, and Gold, and why.

4. Technical Implementation

Implement or provide working SQL and/or PySpark for at least three DQ rules: duplicate detection, referential-integrity validation, and one additional business rule of your choice. The solution should demonstrate how the implementation could be reusable/configuration-driven rather than hardcoded for individual tables.

5. Exception Handling & Remediation

Explain how failed records would be handled. Cover reject vs. quarantine, critical vs. non-critical rules, capturing failed records and failure reasons, and reprocessing corrected records.

6. Reconciliation

Design reconciliation between the source system and the enterprise data platform. The solution should not rely only on total record counts; explain the additional controls you would implement.

7. Monitoring

Propose a simple DQ monitoring approach covering overall DQ score, DQ score by dimension, failed rules, failed records, DQ trends, and critical issues.

8. Root-Cause Analysis

Yesterday, the source contained 1,000,000 enrollment records. Today, Bronze contains 1,000,000 records, Silver contains 970,000 records, and Gold contains 850,000 records.

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Explain step-by-step how you would investigate the differences and determine whether the missing records represent expected business filtering or a data-quality/pipeline issue.

DQ Rule Matrix – Suggested Format

Rule	DQ Dimension	Layer	Severity	Expected Action

Expected Deliverables

- A short solution document or presentation (maximum 5–7 slides/pages).
- A DQ rule matrix.
- SQL and/or PySpark examples for the requested technical controls.
- A simple architecture/process diagram showing where DQ controls are applied.
- Clearly stated assumptions.

Submission

Please submit the completed case study in PDF, PowerPoint, or Word format, with any SQL/PySpark code included in the document or provided as a separate supporting file. Submission Deadline: The completed case study should be submitted within two (2) working days from the date of receipt of this case study.

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